[redacted]

Q1. What would be output of the following code, Explain your answer: **2x5=10 Marks**

Choice 1

a)
```
import sys
def check(x):
    if x % 2 == 0:
        sys.exit()
    return x * 2
print(check(5) + check(3))
```
(CO1)

b)
```
x = 10
y = 20
z = (x < y) and (y > 15) or (x == 10) and not (y ==
20)
print(z)
```
(CO1)

c)
```
def func(a, b=5):
    return a * b
x = func(3)
y = func(2, 10)
print(x + y)
```
(CO1)

d)
```
def compute(x, y=5, z=10):
    return x * y + z
result1 = compute(2)
result2 = compute(3, 4)
result3 = compute(1, z=20)
print(result1, result2, result3)
```
(CO2)

e)
```
word_list = ["hello", "world", "python"]
def modify_word():
    global word_list
    word_list[1] = word_list[1].upper()
    local_list = word_list[:2]
    return "-".join(local_list)
print(modify_word())
print(word_list)
```
(CO2)

OR

Choice 2

a)
```
x = "Automate"
y = "Python"
z = (x[:4] + y[-3:]) * 2
print(z[::3] + z[-2] + z[5])
```
(CO1)

b)
```
def test(n):
    if n % 2 == 0:
        return "Even"
    elif n % 3 == 0:
        return "Divisible by 3"
    return "Odd"
print(test(9) + " " + test(12))
```
(CO1)

c)
```
import pprint
data = {'name': 'Alice', 'age': 25, 'city': 'New York'}
pprint.pprint(data)
```
(CO2)

d)
```
d = {1: 'one', 2: 'two', 3: 'three'}
keys_list = list(d.keys())[::-1]
values_list = list(d.values())[::-1]
print(keys_list[0], values_list[1])
```
(CO2)

e)
```
data = {'Alice': 25, 'Bob': 30, 'Charlie': 35}
try:
    age = data['David']
except KeyError:
    age = sum(data.values()) // len(data)
print(age)
```
(CO2)

Q2. a. Explain the difference between local and global scope in Python. How does the ***global*** statement affect variable scope? Provide an example where modifying a global variable inside a function leads to unexpected behavior.	**3 Marks (CO2)**

SIX SEMESTER [B. TECH.] JAN 2024

Paper Code: CIE-332T | **Subject: Programming in Python**

Time: 1Hour 30 Min. | **Maximum Marks: 30**

Note: Attempt three questions in all including Q. No. 1 which is compulsory.

Q1. (a)	What is the difference between logic error and syntax error?	[2]	CO1
(b)	What are the features of Python Programming language?	[2]	CO1
(c)	Write a Python program to replace last value of tuples in a list. Sample list: [(10, 20, 40), (40, 50, 60), (70, 80, 90)] Expected Output: [(10, 20, 100), (40, 50, 100), (70, 80, 100)]	[2]	CO3
(d)	Create a 5x 2 integer array from a range between 100 to 200 such that the difference between each element is 10.	[2]	CO2
(e)	What is lamda function? Explain with example.	[2]	CO2
Q2. (a)	Write a program to calculate GCD of two numbers.	[5]	CO2
(b)	What are the different types of decision control statements? Explain with examples.	[5]	CO1
Q3. (a)	Write a Python program to calculate the number of minutes in a week. Make variables for DaysPerWeek, HoursPerDay, and MinutesPerHour.	[5]	CO2
(b)	What will be the output of the following Python code snippet? for i in ''.join(reversed(list('abcd'))): print (i)	[2]	CO2
(c)	What is the difference between List, Tuples and Dictionary? Explain in detail.	[3]	CO3
Q4. (a)	Write a program to demonstrate the concept of exception handling in python.	[5]	CO4
(b)	Justify the statement: "Strings are immutable". Explain string manipulation in python programming with the help of code.	[5]	CO3

Date:- 10/6/2024

(Please write your Exam Roll No.) Exam Roll No......................

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JUNE 2024

Paper Code:CIE-332T/IOT-330T	Subject- Programming in Python
Time: 3hours	Max. Marks: 75

Note : Attempt five questions in all including Q. No. 1 which is compulsory. Select one Question from each Unit. Assume suitable missing data if any.

Q1 Answer following in brief. **[5x5 = 25]**

(a) What is the role of data frame? How can data frames be merged?
(b) What are the features of Python Programming language?
(c) Differentiate between list, tuple and dictionary with example.
(d) Write a short note on different methods to read data from a file.
(e) How is exception handling implemented in python?

UNIT-I

Q2 (a) What are loop interruption statements? What is the difference between break, continue and pass? 17. Write a program that prints all integers lying between 1 and 50 that aren't divisible by either 2 or 3. **[6]**
(b) Explain the different numeric, assignment, augmented operators used in python. **[3]**
(c) Explain how operator precedence is used in evaluating expressions. **[3.5]**

Q3 (a) What are the commonly used operators in python? Explain operator overloading with the help of an example. **[6]**
(b) What is type conversion? Write a python program to demonstrate number type conversions. **[6.5]**

UNIT-II

Q4 (a) What are the different ways of string manipulation in python? WAP in python check whether the string is Symmetrical or Palindrome **[6]**
(b) What are functions in python? WAP in python to use Pythagoras theorem to calculate the length of the hypotenuse of a right-angled triangle where the other sides have lengths 3 and 4. **[6.5]**

Q5 (a) Define the following: find(), isalnum(), zfill(), strip(), split() and rindex(). **[6]**
(b) Write the definition of a function Reverse(X). WAP to display the elements in reverse order such that each displayed element is twice of the original element (element *2) of the List X. **[3]**
(c) Write a function mul ( ) which accepts list L, odd_L, even_L as its parameters. Transfer all even values of list L to even_L and all odd values of L to odd_L. **[3.5]**

UNIT-III

Q6 (a) Write a program that accepts filename as an input from the user. Open the file and count the number of times a character appears in the file. **[6]**
(b) What are files? Why do we need them? What are different access modes in which you can open file? **[6.5]**

Q7 (a) What is directory access? How do you walk through a directory tree? **[6]**
(b) Write a python program to merge two files and then change the content of this file to upper case letters. **[6.5]**

P.T.O.

(Please write your Enrolment No. immediately)

Enrolment No. ________

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

VIth Semester, April, 2024

Paper Code: CIE-332T **Subject: Programming in Python**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO's
1 (a)	Discuss the various features of python.	(2)	CO1
(b)	Difference between mutable and immutable objects in python.	(2)	CO1
(c)	What is slicing in list? Explain with example.	(2)	CO3
(d)	Discuss zip () function with an example.	(2)	CO3
(e)	Illustrate the difference between list and tuple?	(2)	CO3
2 (a)	Write Python code to sort a sequence of names according to their alphabetical order without using sort () function.	(5)	CO2
(b)	Illustrate ***args** and ****kwargs** parameters in Python programming language with an example.	(5)	CO2
3 (a)	Explain various data types in python.	(5)	CO1
(b)	Write Python program to check for the presence of a key in the dictionary and find the sum all its values.	(5)	CO3
4 (a)	Discuss the ord(), hex(), oct(), complex() and float() type conversion functions with examples.	(5)	CO1
(b)	Write Python program to implement Stack operations.	(5)	CO2

(Please write your Exam Roll No.) Exam Roll No. 31/5/25

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] MAY-JUNE 2025

Paper Code: CIE-332T/IOT-330T **Subject: Programming in Python**

Time: 3 Hours **Maximum Marks:75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt **any five** of the following: **(5x5=25)**
a) Touple
b) Elements of Flow control
c) Return type
d) Exception Handling
e) Decision trees
f) Parameter

UNIT-I

Q2 a) What is Python Programming? Explain Python features in details. **(6)**
b) Explain different types of data types in Python with example. **(6.5)**

Q3 a) What are operators? Explain types of operators in Python in details **(6)**
b) Write a program in Python to produce Fibonacci series. **(6.5)**

UNIT-II

Q4 a) Explain functions, parameters & types of functions in details with example. **(6)**
b) What is List in Python? Explain in details with example. **(6.5)**

Q5 a) Explain String & its different functions or methods in details with example. **(6)**
b) Write a Python Program to create a simple calculator using functions. **(6.5)**

UNIT-III

Q6 What is the use of Files? Explain reading and writing process on file with an example. **(12.5)**

Q7 a) How can you create Python file that can be imported as a binary as well as run as a standard script? **(6)**
b) What are File input and output operations in Python Programming? **(6.5)**

UNIT-IV

Q8 a) What is module? Explain advantages of modules in details with example. **(6)**
b) Explain Built in Modules in details. **(6.5)**

Q9 a) How to create and import a module in Python. **(6.5)**
b) Explain html, web browser module, MAPIT.PY in details **(6)**

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P-1|1

(Please write your Enrolment No. immediately) Enrolment No. __________

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, March, 2025

Paper Code: CIE-332T **Subject:** Programming in Python

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO mapping
1 a)	Explain the role of break, continue and pass in programming.	(2)	CO1
b)	Which of the following code blocks print- "Simran talks to Ramesh and Suresh"? i. print("Simran talks to Ramesh and Suresh") ii. print("Simran talks to " + "Ramesh"+" and", "Suresh") iii. print("Simran talks to Ramesh", "and", "Suresh") iv. print("Simran talks to" + "Ramesh and Suresh")	(2)	CO1
c)	Suppose dict is a dictionary, dict={"Mohan": 90, "Suresh":80, "Ram": 70, "Suhel":60} What will be the output of following operations- i. dict.items() ii. dict.get("Suresh")	(2)	CO2
d)	Discuss the exit commands used in Python.	(2)	CO1
e)	Distinguish between Mutable and Immutable data types.	(2)	CO1
2 a)	Explain different data types in Python in detail.	(3)	CO1
b)	Write a program to print all prime numbers that fall between two numbers.(Accept two numbers from the user)	(3)	CO1
c)	Develop the following pattern using for loop 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	(4)	CO1
3 a)	Design a function to determine whether the number is an Armstrong number.	(3)	CO2
b)	Consider the following string mySubject: mySubject="Computer Science" What will be the output of the following string operations: i. print(mySubject[-7:-1]) ii. print(mySubject[: : 2]) iii. print(mySubject.isalpha())	(3)	CO2
c)	Write a function that returns the largest element of the list passed as parameter.	(4)	CO2
4 a)	Define scope of a variable in a function. Differentiate between local and global variables with examples.	(3)	CO2
b)	Explain the purpose of *try, except* and *else* blocks in Exception handling.	(3)	CO2
c)	Write a Python program that accepts a string and calculates the number of digits and letters.	(4)	CO2

Please write your Enrolment No. immediately)

Enrolment No. __________

SUPPLEMENTARY MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, April - 2025

Paper Code: CIE-332T **Subject: Programming in Python**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO mapping
1 a)	Write a program in Python that will print the integers from 1 to 50 which are not divisible by 2 or 3.	(2)	CO1
b)	Distinguish between Mutable and Immutable data types.	(2)	CO1
c)	Write a Python program to add a new (key, value) pair to a dictionary only if the key does not exist already.	(2)	CO2
d)	Define loop interruption statements and distinguish among them.	(2)	CO1
e)	Write a function that returns the largest element of the list passed as a parameter.	(2)	CO2
2 a)	Explain different data types in Python with suitable examples.	(5)	CO1
b)	Develop the following pattern using for loop A A B A B C A B C D A B C D E	(5)	CO1
3 a)	Write a program using function to display the elements in reverse order of a list such that each displayed element is twice of the original element of the list.	(5)	CO1
b)	Explain different numeric, assignment, augmented operators used in Python.	(5)	CO1
4 a)	Consider the following string mySubject: mySubject="Computer Science" What will be the output of the following string operations: i. print(mySubject[-8:-2]) ii. print(mySubject[: : -1]) iii. print(mySubject.isalpha())	(5)	CO2
b)	What are different ways of string manipulation in Python. Write a program to check whether the string is palindrome or not.	(5)	CO2